

SUMIT GUPTA

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PROFESSIONAL SUMMARY

Computer Science undergraduate at Vellore Institute of Technology (B.Tech, 2027) focused on systems and machine learning, with hands-on experience building LLM inference servers, distributed systems with Raft consensus, real-time computer vision pipelines, and fairness-audited ML risk models. Python, C++, and Go developer. Increased data pipeline throughput by 35% during a Software Engineering/Data Science internship at SAIL.

TECHNICAL SKILLS

Languages:	Python, C++, Go, TypeScript, JavaScript, C, SQL
Machine Learning & AI:	PyTorch, CUDA, TensorRT, ONNX Runtime, Hugging Face, LLM Inference, Speculative Decoding, PagedAttention, Computer Vision, OpenCV, YOLO, ByteTrack, SHAP, LIME, Whisper, Model Calibration, Fairness Auditing
Systems & Distributed:	Distributed Systems, Raft Consensus, Write-Ahead Log (WAL), Multithreading, Concurrency, Memory-Mapped I/O, Inter-Process Communication, gRPC, WebSockets, WebRTC, TCP/IP Sockets
Backend & Tools:	FastAPI, Streamlit, Node.js, Express.js, Next.js, REST APIs, PostgreSQL, SQLite, Prisma ORM, Redis, Docker, Git, GitHub, Linux
Core Concepts:	Data Structures and Algorithms, Operating Systems, Database Systems, Performance Optimization, Benchmarking

WORK EXPERIENCE

Software Engineering / Data Science Intern, SAIL (Steel Authority of India Ltd.)	Bokaro, India June 2024 – August 2024
- Built and benchmarked Python data extraction pipelines for high-frequency industrial telemetry, increasing end-to-end processing throughput by 35% - Automated data integrity validation across cross-departmental datasets, reducing manual auditing effort by 40% - Refactored legacy SQL queries and data models with senior engineers, eliminating database ingestion bottlenecks around large operational logs	

PROJECTS

Adaptive Speculative LLM Serving Engine	Python, PyTorch, CUDA, FastAPI, Llama-3.1-8B, Qwen2.5 0.5B
- Built a local LLM inference server with entropy-guided dynamic draft lengths (SVIP), stopping speculative decoding early when token confidence drops and reducing wasted compute. - Implemented an $O(1)$ KV-cache rollback mechanism for an 8GB VRAM setup, reusing allocated memory to avoid fragmentation and repeated tensor reallocation. - Achieved a relatively higher token generation throughput than the autoregressive baseline while preserving the target model's output through lossless verification.	
ReadmitIQ: 30-Day Hospital Readmission Risk Prediction with Calibration and Fairness Audit	Python, FastAPI, Streamlit
- Built a readmission risk model on a 69,987-patient cohort (8.99% baseline readmission rate), reaching 0.1385 test PR-AUC. - Captured 34.5% of readmissions in the top 20% highest-risk patients (1.73× lift) with calibrated probabilities (ECE 0.0062). - Audited fairness across race and applied Equalized Odds mitigation, cutting the false-positive-rate gap from 5.07% to 0.17% and the true-positive-rate gap from 6.10% to 3.85%.	
AegisVision: Real-Time Multi-Stream Vision Pipeline	C++, Python, TensorRT, OpenCV, WebRTC
- Built a video analytics engine processing 6 concurrent 1080p RTSP streams at 60+ FPS, using zero-copy shared-memory ring buffers to reduce Python GIL overhead and avoid frame drops. - Used TensorRT FP16 dynamic batching on an NVIDIA RTX 4060 GPU to accelerate YOLO object detection; added ByteTrack tracking and geofencing for safety zone intrusion detection. - Added WebRTC (aiortc) for video streaming and WebSocket telemetry, achieving under 80ms glass-to-glass latency.	
KhorosLog: Distributed Fault-Tolerant Commit Log	Go, Raft, TCP, mmap
- Built a distributed write-ahead log and event streaming engine in Go using Raft consensus, with randomized election timers and automatic leader failover under 150ms. - Added memory-mapped file I/O and a binary sparse index, reaching 80,000+ durable writes per second in benchmarks. - Designed a custom binary wire protocol over TCP with CRC32 checksums, achieving relatively higher serialization throughput of JSON over HTTP in benchmarks.	

EDUCATION

ACHIEVEMENTS & ACTIVITIES

- Solved 150+ data structures and algorithms problems on LeetCode and CodeChef, covering graphs, dynamic programming, and concurrency
- Member, Event Management Club, VIT: coordinated logistics and staging for campus events with 500+ attendees